

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

**Time Duration: 1 Hour
Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

*I M. Arshini / 20424322 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: M 'Arshini'

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Annexure A

Student 15: M ARSHINI | Register Number: 192424322

Question 1 - Library Late Fine

Knowledge Base

1. If a book is returned late, then a fine is charged.
2. The book was returned late.

Goal

Prove that **a fine is charged**, using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 2 - Flight Delay Notification

Knowledge Base

1. If a flight is delayed, then passengers are notified.
2. Flight 202 is delayed.

Goal

Prove that **passengers are notified**, using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 3 - Employee Certification

Knowledge Base

1. If an employee completes the training, then the employee receives a certification.
2. Kevin completed the training.

Goal

Prove that **Kevin receives a certification**, using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 4 - Medical Attention Alert

Knowledge Base

1. If a patient has a fever, then the patient needs medical attention.
2. Ravi has a fever.

Goal

Prove that **Ravi needs medical attention**, using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 5 - Exam Hall Ticket Eligibility

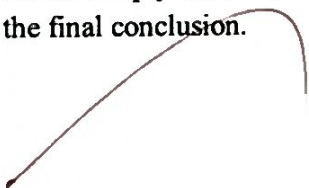
Knowledge Base

1. If a student completes all assignments, then the student is eligible for the exam.
2. If a student is eligible for the exam, then the student receives a hall ticket.
3. The student completed all assignments.

Goal

Prove that **the student receives a hall ticket**, using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
 2. Convert all implications into CNF.
 3. Negate the goal.
 4. Apply the Resolution Algorithm to derive the Empty Clause (Box).
 5. Explain each resolution step and write the final conclusion.
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Library late fine:

Given knowledge Base

* If a book is returned late, then fine is charged

* The book was returned late.

Step 1: propositional logic representation

Let:

L = The book is returned late

F = A fine is charged

statement 1:

$$L \rightarrow F$$

statement 2:-

$$L \wedge \text{Goal} = F$$

Step 2: Convert into CNF

$$P \rightarrow Q = \neg P \vee Q$$

therefore

$$L \rightarrow F$$

becomes

$$\neg L \vee F$$

The second statement is already in CNF:

L

Step 3: negate the goal

$$\text{Goal} = F$$

negated goal:

$$\neg F$$

so the clauses are:

$$C1 = \neg L \vee F$$

$$C2 = L$$

$$C3 = \neg F$$

Step 4: Apply resolution

Resolve c_1 and c_2

$\{L \vee F\}$

(L)

New clause:

$c_4 = F$

Now, resolve c_4 with the negated goal c_3 :

$\{F\}$

$\{\neg F\}$

Step 5: Empty clause:-

the assumption that the goal is false leads to a contradiction.

2) flight delay notification

Given knowledge Base

1) If a flight is delayed, then passengers are notified

2) flight 202 is delayed.

Step 1:- propositional logic representation.

Let :- $D = \text{flight-202 is delayed}$

$N = \text{passengers are notified.}$

statement 1:

$$D \Rightarrow N$$

statement-2:

$$D$$

Goal:

$$N$$

Step 2:- convert into CNF

The implication

$$D \Rightarrow N$$

becomes:

$$\neg D \vee N$$

The fact is

$$D$$

CNF clauses are:

$$C1 = \neg D \vee N$$

$$C2 = D$$

Step 3: Negate the goal

$$\text{Goal} = N$$

Negated goal:

$$\neg N$$

$$C1 = \neg D \vee N$$

$$C2 = D$$

$$C3 = \neg N$$

Step 4: Apply Resolution:

Resolve $C1$ with $C2$:

$$(\neg D \vee N)$$

$$(D)$$

therefore

$$C4 = N$$

Now, resolve $C4$ with the negated goal

$$\begin{array}{l} (N) \\ (\neg N) \end{array}$$

Step 5: empty clause

The empty clause $\square$ is obtained

employee certification:-

- 1) If an employee completes the training then the employee receives a certification
- 2) Kevin completed the training

Propositional Logic representation

Let
 T = Kevin completed the training
 C = Kevin receives certification

Statement 1:

$$T \rightarrow C$$

Statement 2:

$$T$$

Goal:

$$C$$

Step 2: Convert into CNF

Convert the implication

$$T \rightarrow C$$

using:

$$P \rightarrow Q = \neg P \vee Q$$

Therefore;

$$\neg T \vee C$$

The second statement is

$$T$$

CNF clauses:

$$C_1 = \neg T \vee C$$

$$C_2 = T$$

Step 3: Negate the goal

Goal:

$$C$$

Negate the goal: $\neg C$

SD:

$$C_1 = \neg T \vee C$$

$$C_2 = T$$

$$C_3 = \neg C$$

Step 4: - Apply Resolution

Resolution C_1 and C_2 :

$$(\neg T \vee C)$$

$$(T)$$

Therefore:

$$C_4 = C$$

Now resolve C_4 with the negated goal:

$$(C)$$

$$(\neg C)$$

Step 5: Empty clause :-

The resolution process production is empty clause ($\square$)

Medical Attention Alert:

Given knowledge Base

1) If a patient has a fever, then the patient needs medical attention

2) Ravi has a fever

Step 1:- Propositional Logic Representation

let

F = Ravi has a fever

M = Ravi needs medical attention

Statement 1:-

$$F \rightarrow M$$

Statement 2:

$$F$$

Goal:

$$M$$

Step 2:- convert into CNF

convert:-
 $F \rightarrow M$

$$\neg M \vee F$$

The second statement remains F

Therefore:

$$C1 = \neg F \vee M$$

$$C2 = F$$

Step 3: Negate the Goal

$$\text{Goal: } M$$

$$\text{Negated goal: } \neg M$$

compute clause set:

$$C1 = \neg F \vee M$$

$$C2 = F$$

$$C3 = \neg M$$

Step 4:- Apply Resolution

Resolve $C1$ and $C2$:

$$(\neg F \vee M)$$

$$M$$

$$\text{Therefore: } C4 = M$$

Now, resolve $C4$ and $C3$:

$$\begin{array}{c} M \\ (\neg M) \end{array}$$

step 5:

Empty clause:-

The empty clause ($\square$) is obtained.

5) Exam Hall ticket eligibility:

Given:- If a student completes all assignments, then the student is eligible for the exam.

If a student completed all assignments

Propositional Logic representation:-

Let

A = student completed all assignments

E = student is eligible for the exam

H = student receives a hall-ticket

Therefore:

Statement 1:

$$A \rightarrow E$$

Statement 2:

$$E \rightarrow H$$

Statement 3:

Goal:

H

A

Step 2:-

Convert all implications into CNF

For the first implication

$$A \rightarrow E$$

using $p \rightarrow q = \neg p \vee q$

Therefore

$$\neg A \vee E$$

for the second implication

$$E \rightarrow H$$

Therefore

$$\neg E \vee H$$

The third statement is already a clause: A

So, the CNF clauses are:

$$C1 = \neg A \vee E$$

$$C2 = \neg E \vee H$$

$$C3 = A$$

Apply

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	19
Method selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	19
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	24
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	19
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	13

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